



BUS-1023 Applied Mathematics

CLO 2.2

Chain Discount and Cash Discounts

Monday, September 21, 2026



CHAIN or SERIES DISCOUNTS

→ *Discounts given by manufacturers for greater flexibility, like during seasonal trends, promotional activities, etc.*

Example :

The list price of the office equipment is **\$15,000**. The chain discount is **20/15/10**. Calculate the net price.

This refers to 20%, 15%, and 10% discounts given consecutively



CHAIN or SERIES DISCOUNTS

Example: (Solution)

The list price of the office equipment is **\$15,000**. The chain discount is **20/15/10**. Calculate the net price.

Step 1	Step 2	Step 3	Step 4
\$15,000	\$15,000	\$12,000	\$10,200
$\times .20$	$- 3,000$	$- 1,800$	$- 1,020$
$\hline$	$\hline$	$\hline$	$\hline$
\$ 3,000	\$12,000	\$10,200	\$ 9,180 net price
	$\times .15$	$\times .10$	
	$\hline$	$\hline$	
	\$ 1,800	\$ 1,020	



CHAIN or SERIES DISCOUNTS

Example: (Alternate Solution)

The list price of the office equipment is **\$15,000**. The chain discount is **20/15/10**. Calculate the net price and trade discount amount.

$$\text{Net Price} = \text{List Price} \times \text{Net Price equivalent rate}$$

Step 1) Calculate the complement of each rate and convert to decimal.

$$100\% - 20\% = 80\% = 0.8 \quad 100\% - 15\% = 85\% = 0.85 \quad 100\% - 10\% = 90\% = 0.9.$$

Step 2) Calculate the net price equivalent rate. $0.8 \times 0.85 \times 0.9 = 0.612$.

Step 3) Calculate the **net price**. $15,000 \times 0.612 = \$ 9,180$

Step 4) Calculate the **trade discount amount**.

$$15,000 - \$ 9,180 = \$ 5,820$$



CHAIN or SERIES DISCOUNTS

Calculating Trade Discount Amount Using Single Equivalent Discount Rate

Step 1: Find the complement for each trade discount rate.

Complement = 100% - trade discount rate (convert to decimal)

Step 2: Multiply the complements to find the ***net price equivalent rate***.

Step 3: **Single Equivalent Discount rate** = 1 - net price equivalent rate.

Step 4: Trade discount amount = List price x single equivalent discount rate

$$1 - 0.612 = 0.388$$

$$\text{Discount} = 0.388 \times 15,000 = 5,820$$



Example:

Lamps Outlet bought a shipment of lamps from a wholesaler. The total list price was \$12,000 with a 5/10/25 chain discount. Calculate the trade discount amount using single equivalent discount rate.

1. Calculate the complement of each chain discount.

$$\begin{array}{r} 100\% \\ - 5 \\ \hline 95\% \end{array}$$

$$\begin{array}{r} 100\% \\ - 10 \\ \hline 90\% \end{array}$$

$$\begin{array}{r} 100\% \\ - 25 \\ \hline 75\% \end{array}$$

2. Calculate the net price equivalent rate.

$$\text{NPER} = 0.95 \times 0.90 \times 0.75 = 0.64125$$

3. Calculate the net price.

$$\text{Net price} = 0.64125 \times 12,000 = 7,695$$

4. Calculate the single equivalent discount rate.

$$\text{SEDR} = 1 - 0.64125 = 0.35875$$

5. Calculate the trade discount amount.

$$\text{Discount} = 0.35875 \times 12,000 = \$ 4,305$$



Practice:

A company purchases office equipment with a listed price of **AED 12,500**. The supplier offers series discounts of **10%**, **5%**, and **4%**.

Calculate:

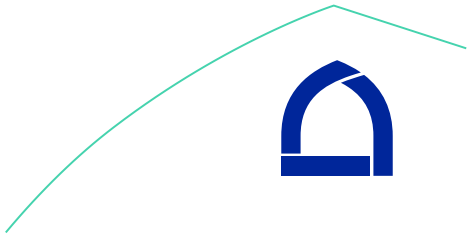
- a. The final price after all three discounts.
- b. The total amount of discount received.
- c. The **single equivalent discount rate** that would give the same final price.



Practice:

A retail company purchases **computer equipment** with a listed price of **AED 15,000**. The supplier offers successive series discounts of **12%, 6%, and 3%**.

Calculate:

- a. The **final price** after applying all three discounts.
 - b. The **total amount of discount** received.
 - c. The **single equivalent discount rate** that would result in the same final price.
- 



Invoice – Cash Discount

New Hampshire Propane Company				
Date	Description	Qty.	Price	Total
06/24/16	Previous Balance			\$0.00
	PROPANE	3.60	\$3.40	\$12.24
Invoice No.		Totals this invoice:		\$12.24
004433L		AMOUNT DUE:		\$12.24
Invoice Date		Prompt Pay Discount:		\$0.19
6/26/16		Net Amount Due if RECEIVED by 07/10/16:		\$12.05
Due Date		7/26/16		





A cash discount

- is a reduction in the amount a buyer must pay when the buyer **pays the invoice early**, within a specified discount period.



Cash Discounts

Credit Period

Mar. 1

Mar. 31

Time period sellers give buyers to pay invoices.

Discount Period

Mar. 1

Mar. 10

Time period buyer has to take advantage of cash discount.



Calculating Credit & Discount Periods

Days-in-a-Year Calendar

Exact days-in-a-year calendar (excluding leap year)*

Day of month	31 Jan.	28 Feb.	31 Mar.	30 Apr.	31 May	30 June	31 July	31 Aug.	30 Sept.	31 Oct.	30 Nov.	31 Dec.
1	1	32	60	91	121	152	182	213	244	274	305	335
2	2	33	61	92	122	153	183	214	245	275	306	336
3	3	34	62	93	123	154	184	215	246	276	307	337
4	4	35	63	94	124	155	185	216	247	277	308	338
5	5	36	64	95	125	156	186	217	248	278	309	339
6	6	37	65	96	126	157	187	218	249	279	310	340
7	7	38	66	97	127	158	188	219	250	280	311	341
8	8	39	67	98	128	159	189	220	251	281	312	342
9	9	40	68	99	129	160	190	221	252	282	313	343
10	10	41	69	100	130	161	191	222	253	283	314	344
11	11	42	70	101	131	162	192	223	254	284	315	345
12	12	43	71	102	132	163	193	224	255	285	316	346
13	13	44	72	103	133	164	194	225	256	286	317	347
14	14	45	73	104	134	165	195	226	257	287	318	348
15	15	46	74	105	135	166	196	227	258	288	319	349
16	16	47	75	106	136	167	197	228	259	289	320	350
17	17	48	76	107	137	168	198	229	260	290	321	351
18	18	49	77	108	138	169	199	230	261	291	322	352
19	19	50	78	109	139	170	200	231	262	292	323	353
20	20	51	79	110	140	171	201	232	263	293	324	354
21	21	52	80	111	141	172	202	233	264	294	325	355
22	22	53	81	112	142	173	203	234	265	295	326	356
23	23	54	82	113	143	174	204	235	266	296	327	357
24	24	55	83	114	144	175	205	236	267	297	328	358
25	25	56	84	115	145	176	206	237	268	298	329	359
26	26	57	85	116	146	177	207	238	269	299	330	360
27	27	58	86	117	147	178	208	239	270	300	331	361
28	28	59	87	118	148	179	209	240	271	301	332	362
29	29	—	88	119	149	180	210	241	272	302	333	363
30	30	—	89	120	150	181	211	242	273	303	334	364
31	31	—	90	—	151	—	212	243	—	304	—	365



Use the "Days-in-a-Year" Calendar to calculate the following **credit due dates**:

a) 90 days from August 12th

b) 80 days from December 5th



Scan the QR-code using your smart phone camera. You will be prompted to open a link in Chrome (or any other browser). You will see a calendar.





Use the “Days-in-a-Year” Calendar to calculate the following **credit due dates**:

a) 90 days from August 12th

August 12 refers to **224**

$$224 + 90 = 314$$

314 Refers to **November 10**

	Exact days(excluding leap year)				
Days of month	31	30	31	30	31
	Aug	Sep	Oct	Nov	Dec
1	213	244	274	305	335
2	214	245	275	306	336
3	215	246	276	307	337
4	216	247	277	308	338
5	217	248	278	309	339
6	218	249	279	310	340
7	219	250	280	311	341
8	220	251	281	312	342
9	221	252	282	313	343
10	222	253	283	314	344
11	223	254	284	315	345
12	224	255	285	316	346
13	225	256	286	317	347



Use the "Days-in-a-Year" Calendar to calculate the following **credit due dates**:

b) 80 days from **December 5th**

(1) Read the number from the calendar for the date 5 and month Dec.
This number is = **339**

(2) Add $339 + 80 = \mathbf{419}$
This number is greater than 365.
So our expected answer will be a date from the next year.

(3) Subtract $419 - 365 = \mathbf{54}$. Find the date for this number.

(4) Expected date is **23-Feb (next year)**

Days of month	31	28	31
	Jan	Feb	Dec
1	1	32	335
2	2	33	336
3	3	34	337
4	4	35	338
5	5	36	339
6	6	37	340
7	7	38	341
8	8	39	342
9	9	40	343
10	10	41	344
11	11	42	345
12	12	43	346
13	13	44	347
14	14	45	348
15	15	46	349
16	16	47	350
17	17	48	351
18	18	49	352
19	19	50	353
20	20	51	354
21	21	52	355
22	22	53	356
23	23	54	357
24	24	55	358
25	25	56	359
26	26	57	360
27	27	58	361
28	28	59	362
29	29	-----	363
30	30	-----	364
31	31	-----	365





Understanding payment terms

(1) Understanding terms of payments **2/10, n/30**

2/10: 2% discount within first 10 days

First number: discount rate 2%
Second number: number of days, 10 from the invoice date

n/30: Credit for 30 days, but no discount if payment made after 10 days. (10 is related to the first term 2/10).

Please do not read 2/10 as chain discount 2% followed by 10%.

(2) Understanding terms of payments **2/10, 1/15, n/30**

2/10: 2% discount within first 10 days

1/15: 1% discount if payment made after 10 days but before 15 days (10 is related to the first term 2/10)

n/30: Credit for 30 days, but no discount if payment made after 15 days. (15 is related to the second term 1/15).



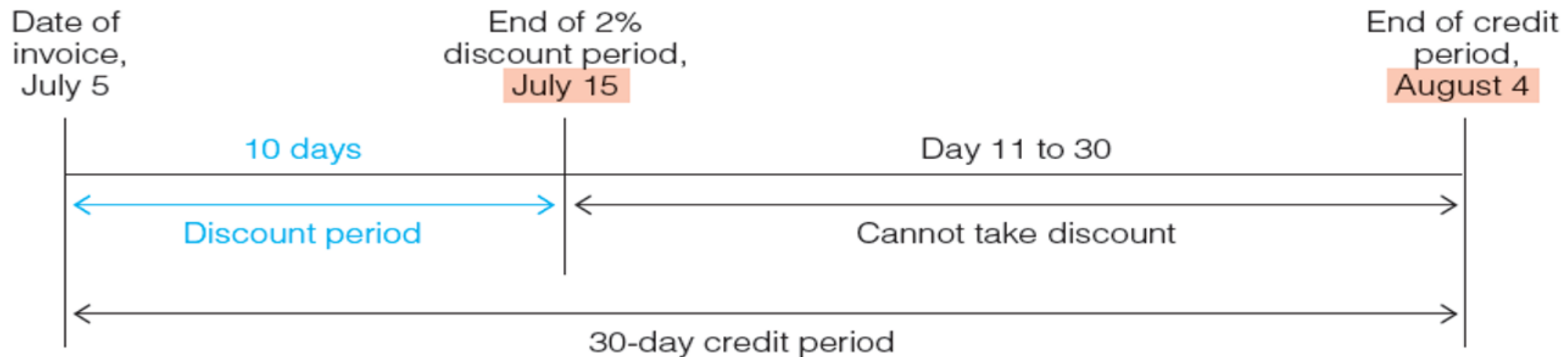


Calculating Cash Discounts

Example 1: \$400 invoice dated July 5; terms 2/10, n/30; paid on July 11.

$\$400 \times .02 = \8 cash discount

$\$400 - \$8 = \$392$ paid OR $\$400 \times .98 = \392





Calculating Cash Discounts

Example 2: \$600 invoice dated **May 8th**. Terms: **2/10, 1/15, n/30**, paid on May **22nd**.

- a) Calculate the end of the 2% discount period.
- b) Calculate the end of the 1% discount period
- c) Calculate the end of the credit period.





Calculating Cash Discounts

Example 2: \$600 invoice dated **May 8th**. Terms: **2/10, 1/15, n/30**, paid on May 22nd.

- a) Calculate the end of the 2% discount period.
- b) Calculate the end of the 1% discount period
- c) Calculate the end of the credit period.
- d) Calculate the amount to be paid, if the payment is made on May 22nd.

Solution:

May 8		<u>10</u> days	<u>15</u> days	<u>30</u> days
128 (from calendar)		128+ <u>10</u> = 138	128+ <u>15</u> = 143	128+ <u>30</u> = 158
Days of month	31	18 th May (Date from calendar)	23 rd May (Date from calendar)	7 th June (Date from calendar)
	May	<i>End of 2% discount period</i>	<i>End of 1% discount period</i>	<i>End of credit period</i>
8	128			



Calculating Cash Discounts

Example 2: \$600 invoice dated **May 8th**. Terms: **2/10, 1/15, n/30**, paid on **May 22nd**.

Solution:

d) Calculate the amount to be paid, if the payment is made on May 22nd.

Find the number of days between May 8th and May 22nd, from the calendar.

Number of days = 142 - 128 = **14 days**.

1% discount is applicable as the payment is made after 10 days but within 15 days.

Cash discount amount = **1% × 600 = 6**

Amount to be paid after cash discount = **600 - 6 = \$ 594**





Practice 1:

An AED 500 invoice dated April 10th has terms of payment 2/10, 1/15, n/30. Find the following:

- a) Discount rate if payment is made within 10 days
- b) Discount rate if payment is made within 12 days
- c) Discount rate if payment is made within 15 days
- d) Amount to be paid if payment is made on April 15th
- e) Discount amount if payment is made on April 22nd
- f) Amount to be paid if payment is made on April 28th





Practice 2:

A store received an invoice dated June 28, 2022, from a manufacturer that sells office equipment. The list price was AED 22,000. This manufacturer offers its customers a trade discount of 15%. The manufacturer also offers its customers the following cash discount terms: 2/10, n/30. Assuming that the store pays the invoice on June 30, 2022, calculate the following.

(Round your answers to the nearest cent.)

- a) Trade discount amount =
- b) Net price =
- c) Cash discount amount =
- d) The amount paid to the manufacturer =



Higher
Colleges of
Technology



كليات
التقنية
العليا

Thank You



800 MyHCT (800 69428)



www.hct.ac.ae

